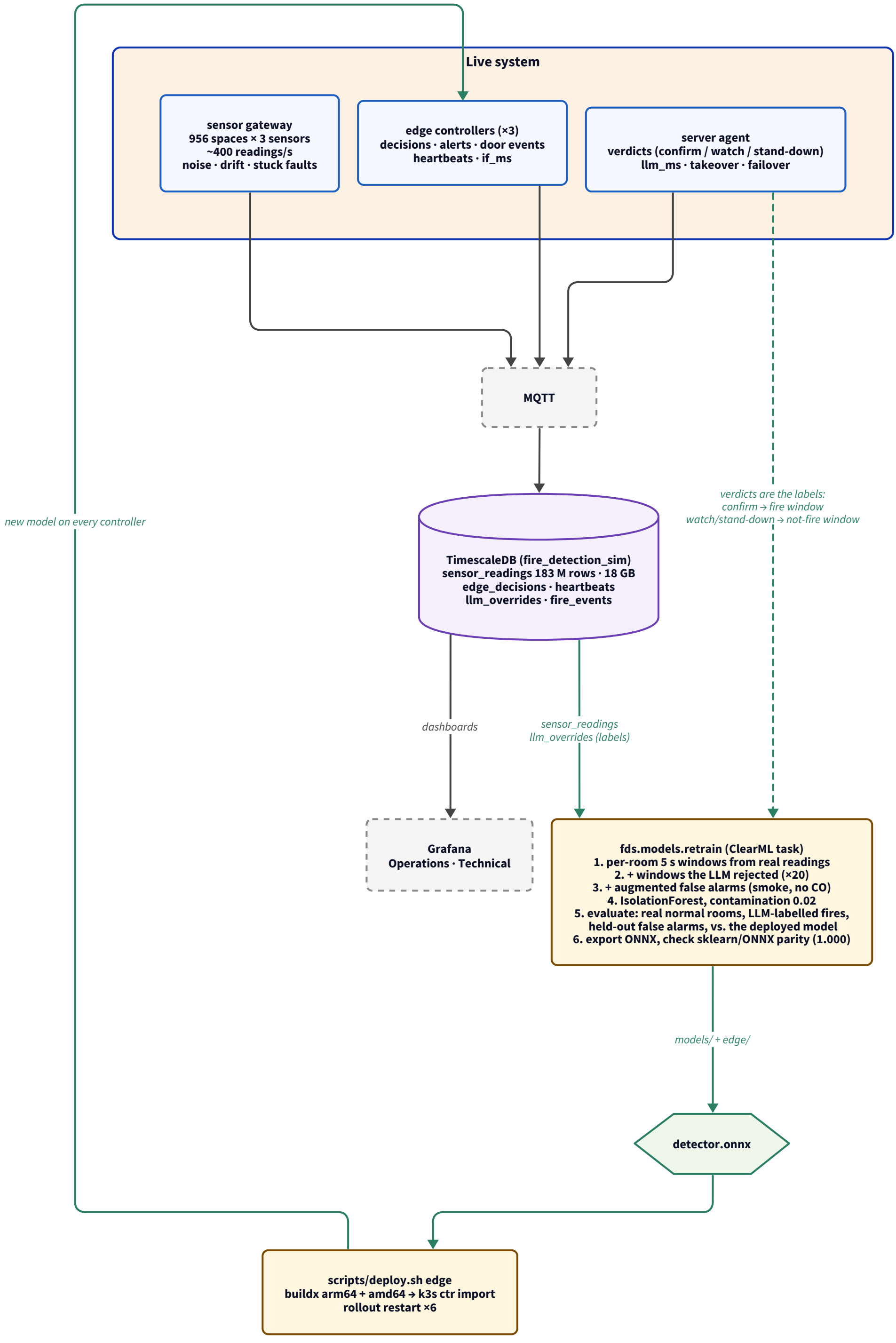

Live system
sensor gateway
956 spaces × 3 sensors
~400 readings/s
noise · drift · stuck faults
edge controllers (×3)
decisions · alerts · door events
heartbeats · if_ms
server agent
verdicts (confirm / watch / stand-down)
llm_ms · takeover · failover
MQTT
TimescaleDB (fire_detection_sim)
sensor_readings 183 M rows · 18 GB
edge_decisions · heartbeats
llm_overrides · fire_events
new model on every controller
verdicts are the labels:
confirm → fire window
watch/stand-down → not-fire window
dashboards
sensor_readings
llm_overrides (labels)
Grafana
Operations · Technical
fds.models.retrain (ClearML task)
1. per-room 5 s windows from real readings
2. + windows the LLM rejected (×20)
3. + augmented false alarms (smoke, no CO)
4. IsolationForest, contamination 0.02
5. evaluate: real normal rooms, LLM-labelled fires,
held-out false alarms, vs. the deployed model
6. export ONNX, check sklearn/ONNX parity (1.000)
models/ + edge/
detector.onnx
scripts/deploy.sh edge
buildx arm64 + amd64 → k3s ctr import
rollout restart ×6